ML Challenge 2025: Smart Product Pricing Solution

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Submission Date: [12-oct-2025]

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1. Executive Summary

We developed a machine learning solution to predict product prices based on textual product descriptions. Our approach combines **TF-IDF features**, **Sentence Transformer embeddings**, and **numeric text features** with a **LightGBM regressor** to model complex relationships between product attributes and pricing.

The model achieves a **validation SMAPE of ~53%**, indicating moderate predictive performance. Future improvements are planned to reduce this further below 35% by adding categorical, numeric, and additional features.

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2. Methodology Overview

2.1 Problem Analysis

- The target `price` has a **wide distribution**, with many low-priced items and a few extremely high-priced items.
- `catalog_content` contains product titles, descriptions, and Item Pack Quantity (IPQ), which are predictive of price.
- No missing values remain in the target column after preprocessing.

2.2 Solution Strategy

- **Approach Type:** Single-model regression with feature engineering.
- **Core Innovation:** Combined **TF-IDF + SentenceTransformer embeddings + numeric text features** (text length, word count, digit count) for improved accuracy.

3. Model Architecture

3.1 Architecture Overview

- **Input:** `catalog_content` text.
- **Features:**
  - TF-IDF (max_features=5000, ngram_range=(1,2))
  - Sentence Transformer embeddings (`all-MiniLM-L6-v2`)
  - Numeric text features: text length, word count, digit count
- **Model:** LightGBM regressor trained on **log-transformed prices**.
- **Output:** Predicted price (after applying inverse log transformation).

3.2 Model Components

- **Text Processing Pipeline:**
  - Preprocessing: Fill missing values.
  - Feature extraction: TF-IDF + Sentence Transformer embeddings.
- **Numeric Features:** Extracted from text (text length, word count, digit count).
- **Key Parameters:**
  - TF-IDF: `max_features=5000`, `ngram_range=(1,2)`
  - SentenceTransformer: batch_size=64
  - LightGBM: `n_estimators=1000`, `num_leaves=64`, `learning_rate=0.03`, `max_depth=12`, `early_stopping_rounds=50`
- **Image Processing Pipeline:** Not used (future improvement possible).

4. Model Performance

4.1 Validation Results

- **SMAPE:** 53.35% (validation set)
- **Notes:** Model underfits slightly due to sparse features and limited numeric/categorical data.

Other metrics (optional):

- RMSE on log-transformed prices: 0.4777

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5. Conclusion

- The pipeline efficiently predicts product prices using **text-based features**.
- Current validation SMAPE indicates room for improvement.
- **Future improvements:**
    - Incorporate categorical and numeric features from product metadata.
    - Include image features if available.
    - Fine-tune embeddings and/or ensemble multiple models to reduce SMAPE below 35%.

Appendix

- **Code artifacts:** https://github.com/piyush-1234567/mlProject
- **Libraries used:** pandas, numpy, sklearn, lightgbm, sentence-transformers, scipy